

Emmanuel Millan

(954) 338-0010 • emmanueljmillan@outlook.com • Miami, FL
linkedin.com/in/emmanueljmillan/

EDUCATION

University of Miami

M.S. in Neural Engineering. GPA: 3.67

Dec 2025

University of Miami

B.S. in Biomedical Engineering, Premed Concentration. GPA: 3.60
Double Major in Mathematics, Applied Analysis Concentration

May 2022

RESEARCH EXPERIENCE

Graduate Research Assistant at NeuSTEM Lab

Nov 2024 – Dec 2025

The Miami Project to Cure Paralysis

Master's Project: *Characterization of Nanoparticle Interactions in Neural Lesion Models*.

- Perform primary Schwann cell isolation from adult rat sciatic nerve, sciatic nerve tissue dissection, explant digestion, culture maintenance, and fluorescent ICC staining with imaging.
- Dissect brain and spinal cord tissue from mice for CNS myelin debris isolation, including homogenization and density gradient centrifugation.
- Develop and conduct nanoparticle treatment studies, performing DMMB, MTS, and BCA assays.
- Fabricate and characterize experimental biomaterials (nanoparticles, microspheres), performing enzymatic depolymerization, ultrafiltration, lyophilization, and calculating zeta-potential.
- Support in vivo wound-healing studies, including assisting in rat surgeries and image processing in Fiji (ImageJ).
- Contribute to scientific writing for lab manuscripts, including co-authoring a manuscript on biomolecular corona-tuned nanoparticle immunomodulation (submitted to Nanomedicine, July 2025).

POSTER PRESENTATIONS

“Characterization of Nanoparticle Interactions in Neural Lesion Models”

University of Miami 2025 Neuroscience Research Day, Miami, FL, December 2025

“Characterization of Schwann Cell-Nanoparticle Interactions in Neural Lesion Models”

University of Miami 2025 College of Engineering Research Symposium, Coral Gables, FL, October 2025

PROFESSIONAL EXPERIENCE

Biomedical Research Engineer

Aug 2022 - Jul 2025

Sigma Biomedical – Sunrise, FL

- Performed device instrumentation and data analysis with oscilloscopes, Doppler ultrasound systems, spectroradiometers, and electrical safety analyzers.
- Built and compiled three MATLAB apps to quantify endoscope performance parameters, such as image contrast, image intensity uniformity, and RGB color accuracy (per ISO 8600-1).
- Designed, validated, and documented protocols for acoustic, optical, and electrical safety testing in accordance with ISO and IEC standards, including IEC 60601-1, ISO 8600-1, IEC 60601-2-37.
- Fabricated tissue-mimicking phantoms and developed closed-loop flow system to simulate physiological vascular conditions for ultrasound catheter testing.
- Supported regulatory affairs and contributed to FDA 510(k) submissions.

Research and Development Intern

Jun 2022 - Dec 2022

Terumo Aortic – Sunrise, FL

- Evaluated stent-graft deployment forces into 3D aorta vasculature model using force transducers.
- Headed changes within QMS such as document creation and revisions, including SolidWorks drawings.
- Performed lab technician duties, including conducting maintenance checks, tracking equipment calibration, and experimental assembly.

SERVICE AND OUTREACH

Hospital Volunteer

July 2024 - Present

Joe DiMaggio Children's Hospital – Hollywood, FL

- Provide bedside companionship and play for pediatric inpatient and support Child Life across units.
- Coordinate patient-advocacy and staff recognition events, handling planning, scheduling, and materials as a member of the Patient-Family Advisory Council

HONORS AND DISTINCTIONS

Dean's List | Fall 2018, Fall 2019, Spring 2022

Provost's Honor Roll | Fall 2019, Spring 2022

CERTIFICATIONS AND RESEARCH COMPLIANCE TRAINING

Certifications

Basic Life Support (BLS) Certification – American Red Cross, 2024.

CITI Program Training

Biomedical Responsible Conduct of Research; Working with the IACUC; Working with Mice in Research; Working with the Laboratory Rat – Sep 2025.

Aseptic Surgery; Reducing Pain and Distress; University of Miami, Office of Health and Safety and SOPs – Aug 2025.

TECHNICAL SKILLS

Software/Computational: MATLAB, Python, Fiji (ImageJ), Arduino, Microsoft Excel, SAS JMP, GraphPad Prism, R, LabVIEW, SolidWorks, SpectraWiz

Wet Lab: ICC, MTS, DMMB, PIC, Primary Cell Isolation, Tissue Dissection, Cell Culture Maintenance, Biomaterial Fabrication, SDS-PAGE

Equipment: Fluorescence Spectroscopy, UV-Vis Spectroscopy, Microscopy (NIS-Elements), Microplate Reading, Electrical Testing, Rheometry, Oscilloscopes, Ultrasound Systems

Regulatory/Standards: ISO 60601-1, ISO 8600-1, QMS, FDA 510(k) Preparation